QP Code: 25901131

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth Semester
Value Addition Course (VAC) - MG4VACBOT200 - Conservation Biology And Sustainable Development
(2024 Admission Onwards)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short answer
Answer any **10** questions. Each question carries **1 marks**

1. Mention one negative impact of mainstream tourism. [U] / [CO1
   Environmental degradation, habitat destruction, loss of biodiversity **(Any One)**
2. Mention one job role related to conservation biology in government or NGO sector.[K] / [CO1]
   Wildlife Biologist, Conservation Biologist, Forest officer, Marine Conservation Specialist. **(Any One)**
3. What are tissue culture banks? [K] / [CO2]
   Tissue culture banks (also known as cell culture banks or bio banks) are specialized facilities that collect, process, and preserve biological materials under strictly controlled conditions for future use in research, medicine, or conservation.
4. State one objective of PVA in conservation biology. [U] / [CO2]
   A core objective of Population Viability Analysis (PVA) is to predict the probability of extinction for a specific population over a set timeframe
5. List out any two objectives of green practices in ecotourism. [K] / [CO1, CO4]
   Environmental Conservation and Protection
   Empowerment of Local Communities
6. How many biodiversity hotspots are found in India? [K] / [CO5]
   Four
7. What is meant by amber colour coding in red data book? [U] / [CO4, CO5]
   In the Red Data Book, amber colour coding represents species categorized as Vulnerable
8. Why are biodiversity hotspots important for RET species? [K] / [CO4, CO5]
   Biodiversity hotspots are critical for RET species because they act as "last refuges" for biological diversity
9. Connect any two SDGs that are related to sustainable forestry. [K] / [CO3, CO6]
   SDG 15 (Life on Land) , SDG 13 (Climate Action).
10. How many categories are there in the IUCN Red List? [K] / [CO4, CO5]

Nine

11. Which are the core elements of sustainable development? [U] / [CO6]
    Environmental Protection, Social Equity, Economic Viability
12. State any two applications of solar energy. [U] / [CO3, CO6]
    Solar Photovoltaics , Solar Thermal Heating

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions Each question carries **5 marks**

13. Discuss the role of afforestation and reforestation in conservation and sustainable development.[K] / [CO1]
    - Climate Change Mitigation
    - Biodiversity Enhancement
    - Soil and Water Conservation
    - Restoring Degraded Land
    - Poverty Reduction and Economy
    - Sustainable Agriculture **(Any Four Points with explanation)**
14. Describe the importance of protecting biodiversity. [U] / [CO2]
    Protecting biodiversity is essential for sustaining life on Earth, as it provides critical ecosystem services like air/water purification, climate regulation, and pollination. It maintains ecological balance, enhances resilience to environmental change, and offers invaluable resources for food, medicine, and economic stability, ensuring long-term human survival
15. Describe the positive effects of tourism on local economies. [K] / [CO1]
    - Economic Diversification
    - Cultural & Environmental Preservation
    - Job Creation
    - Income Growth
    - Infrastructure Development.
    - Support for Local Businesses (Multiplier Effect)
    - Foreign Exchange Earnings **(Any Four Points with explanation)**
16. Discuss the three pillars of sustainable development. [K] / [CO6]
    Sustainable development is based on three main pillars that must be balanced to ensure long-term well-being for people and the planet. These pillars are:
    - Economic Sustainability: Focuses on maintaining economic growth and development without causing harm to the environment or society. Ensures that resources are used efficiently and responsibly to support livelihoods and economic activities over time.
    - Environmental Sustainability: Involves protecting natural resources and ecosystems to maintain the health of the planet. Aims to reduce pollution, conserve biodiversity, and manage natural resources so they are available for future generations.

- Social Sustainability: Emphasizes social equity, justice, and the well-being of communities. Ensures access to basic needs such as education, healthcare, and human rights, promoting inclusive and resilient societies.

17. Discuss conservation methods for endemic species. [U] / [CO4, CO5]
    Endemic species, restricted to specific geographical areas, are highly vulnerable to extinction due to habitat loss, climate change, and invasive species. Conservation methods includes in situ conservation and ex situ conservation.
    (Brief explanation on conservation method with examples)
18. Describe ecosystem services provided by biodiversity. [U] / [CO4]
    - Supporting (e.g. soil formation, nutrient cycling, primary production)
    - Provisioning (e.g. food, fresh water, fuelwood, fiber, biochemicals, genetic resources)
    - Regulating (e.g. climate regulation, disease regulation, water regulation, water purification, pollination)
    - Cultural (e.g. spiritual and religious, recreation and ecotourism, aesthetic, inspirational, educational, sense of place, cultural heritage)

19. Illustrate how green spaces are linked to the sustainable development goals?[An] / [CO3, CO6]
    Green spaces, including parks, forests, and wetlands, are critical to achieving the UN Sustainable Development Goals (SDGs) by improving public health (SDG 3), providing clean air/water (SDG 6/15), and promoting sustainable cities (SDG 11). They act as ecological infrastructure to mitigate climate change, foster social cohesion, and protect biodiversity in urban landscapes.
20. Illustrate the role of water efficient fixtures in attaining SDGs. [An] / [CO3, CO6]
    Water-efficient fixtures (low-flow toilets, faucet aerators, showerheads) reduce indoor water use by up to 55%, directly advancing SDG 6 (Clean Water and Sanitation).
    By lowering demand, they mitigate water scarcity, decrease energy used for water heating/treatment (SDG 13/7), and promote sustainable urban infrastructure (SDG 11).

[5x6 = 30]

Part C

Essay

Answer any **1** questions Each question carries **10 marks**

21. Explain the various human-induced causes of biodiversity loss and their consequences. [K] / [CO4, CO5]
    Causes of biodiversity loss
    - Habitat Destruction & Fragmentation
    - Overexploitation of Resources
    - Climate Change
    - Pollution
    - Invasive Species.

Consequences of Biodiversity Loss

- Ecosystem Collapse & Functional Loss
- Reduced Food Security & Health Risks
- Economic and Social Impact
- Species Extinction

(Cause with explanation 5marks + Consequence 5marks)

22. Describe the strategies and policies for sustainable development. [K] / [CO3, CO6]

Sustainable development principles combine social, economic, and environmental goals to promote global well-being. They aim to provide quality education, healthcare, and employment while preserving ecosystems and protecting all forms of life. These principles are enshrined in the UN's 17 Sustainable Development Goals

- Protecting the natural environment and ecosystems
- Safeguarding global biodiversity and species variety
- Promoting inclusive and long-term societal progress
- Enhancing and preserving human capital through health and education
- Regulating population growth for resource balance

Strategies for Sustainable Development

1. Use of Non-conventional sources for energy
2. Establishing/installing Mini-hydel Plants
3. Cleaner fuel use in both rural and urban regions
4. Utilisation of bio-compost
5. Biopest control
6. Traditional practices and knowledge
7. Awareness among individuals

(Brief explanation with any other relevant points can give full credit)

[10x1 = 10]

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